This guide will help you install the Kathará tool on your computer. Any supported OS (windows, linux, mac os) will work, although linux is encouraged.

Kathará (Kathara from now on for ascii friendliness) is a network emulation tool https://www.kathara.org/. With this tool we can build networks with any topology having any number of devices. Each device is implemented by a linux container and can be configured arbitrarily. We will use Kathara throughout the lab section of the course.

You can **choose one** of two methods to use Kathara, one installing it natively on your computer and the other using a pre-configured virtual machine.

Method 1: Kathara native on your computer (encouraged).

1. **Install docker**. Docker will be the backend container manager for Kathara. If you don't have it already installed on your computer:
    - Follow the steps available here https://docs.docker.com/get-docker/ for your OS. On Windows, during installation procedure, do not flag "Use Windows Containers".

2. **Install Kathara**.
    - Go to https://github.com/KatharaFramework/Kathara/wiki/Installation-Guides to download and install kathara for your OS.

3. **Set our custom docker image for Kathara**. We have built a custom docker image containing all the tools we'll need throughout the course. You have to tell Kathara to use this image instead of the default one. This can be done as follows:
    - Type `kathara settings`
    - Select "Choose default image"
    - Select "Choose another image"
    - Type "`pietrospadaccino/networkinfrastructures`" and enter. If you are on MacOS with Apple Silicon (2020 or later) use instead "`pietrospadaccino/networkinfrastructures_m1`".

4. **Run Kathara checks** (don't skip)**.** This downloads the docker image and runs checks:
    - Type `kathara check`

Method 2: Kathara on a Virtual Machine. Go to the shared folder: https://drive.google.com/drive/folders/1yI7e9glutqr-NN6fqNLgvdAHb-5yBvca?usp=drive_link

- If you are on Windows or older Intel Macs:
    - Install VirtualBox https://www.virtualbox.org/wiki/Downloads
    - Download and import "kathara.ova"
- If you are on Mac with Apple Silicon:
    - Install UTM https://mac.getutm.app/
    - Download, unzip, and import "kathara.utm.zip"

For the virtual machines the user is "user" and password is "user".